Smart Textile Manufacturing Quality Inspection and Production Optimization Platform

RESEARCH PAPER REVIEW/LITERATURE SURVEY

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Submitted by

B Priyanka (192421206)

Under the supervision of

Dr. K. Anithadavamani

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai - 602105

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1. INTRODUCTION

1.1 Background

The textile manufacturing industry requires reliable quality inspection because fabric defects can reduce product quality, increase material waste, and affect production efficiency. Traditional manual inspection is time-consuming and can be affected by human fatigue and inconsistency. Recent research has therefore focused on computer vision, machine learning, and deep learning for automated fabric defect detection.

Deep learning-based inspection has become an important research direction because it can automatically learn visual features from fabric images. Recent studies have investigated YOLO-based object detection, attention mechanisms, transfer learning, anomaly detection, and lightweight models for real-time industrial inspection.

The proposed **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform** extends this area by combining automated defect inspection with machine monitoring, production analysis, alerts, reporting, and optimization. This literature survey reviews existing research related to these technologies and identifies the limitations that should be addressed in an integrated smart textile manufacturing platform.

1.2 Objectives of the Literature Survey

The main objectives are:

- To study existing research on automated textile and fabric defect detection.
- To compare machine learning and deep learning approaches.
- To examine datasets, tools, and evaluation methods used in existing studies.
- To identify limitations of current inspection approaches.
- To identify research gaps related to real-time and intelligent textile manufacturing.
- To suggest future improvements relevant to the proposed platform.

2. SELECTION AND REVIEW OF RESEARCH PAPERS

2.1 Paper Selection Criteria

Research papers were selected based on their relevance to textile manufacturing, fabric defect detection, computer vision, deep learning, real-time inspection, and intelligent quality control. Preference was given to recent peer-reviewed research and established review studies.

The selected literature includes studies published between **2022 and 2025**, covering different approaches such as YOLO-based object detection, transfer learning, anomaly detection, and comprehensive reviews of fabric inspection systems.

2.2 Selected Research Papers

Paper 1: Kahraman and Durmuşoğlu (2023)

Title: *Deep learning-based fabric defect detection: A review*

This paper reviews the development of deep learning approaches for fabric defect detection. The authors examine research published from 2003 onward and discuss how deep learning has increasingly been applied to textile defect inspection. The review provides a broad understanding of the methods used in this research area and identifies important challenges in applying deep learning to fabric inspection.

The paper is useful for understanding the evolution of deep learning-based textile inspection and the major research directions in the field.

Paper 2: Lin et al. (2023)

Title: *An Efficient and Intelligent Detection Method for Fabric Defects based on Improved YOLOv5*

Lin et al. proposed an improved YOLOv5-based method to address small fabric defects, imbalanced defect sizes, and detection-speed limitations. The approach incorporated a Swin Transformer module, weighted bidirectional feature processing, an additional detection layer, and generalized focal loss.

The researchers used the Alibaba Cloud Tianchi textile dataset containing 10,321 images and eight defect categories. The proposed method achieved improved detection performance and was designed with real-time embedded-device requirements in mind.

The main limitation is that the approach is primarily focused on visual defect detection rather than integrating inspection with wider production monitoring and optimization.

Paper 3: Lin et al. (2023)

Title: *Automatic Fabric Defect Detection Method Using AC-YOLOv5*

This study proposed AC-YOLOv5 to handle complex textile backgrounds, different defect sizes, and varying defect types. The method introduced Atrous Spatial Pyramid Pooling (ASPP) and a Convolution Squeeze-and-Excitation (CSE) attention mechanism into YOLOv5. The researchers collected fabric images using an inspection system installed on a circular knitting machine and created a self-built industrial dataset. The reported overall detection accuracy was 99.1%.

The study demonstrates the importance of adapting detection networks to textile-specific image characteristics. However, the dataset is self-built and the solution focuses primarily on defect detection rather than a complete production optimization system.

Paper 4: Szarski and Chauhan (2022)

Title: *An unsupervised defect detection model for a dry carbon fiber textile*

This study investigated unsupervised defect detection for dry carbon textiles used in aerospace manufacturing. The research addresses a major practical problem: obtaining a comprehensive dataset of defective samples can be difficult because defects are relatively rare, while normal production data are much easier to collect.

The research demonstrates the value of unsupervised approaches when labelled defect data are limited. However, the study focuses on a specific carbon-fiber textile application and does not address the wider requirements of multiple textile production processes.

Paper 5: Hu et al. (2024)

Title: *Fabric Defect Detection Method Using SA-Pix2pix Network and Transfer Learning*

Hu et al. proposed an SA-Pix2pix-based approach combined with transfer learning to improve detection of complex patterned fabric defects when training samples are limited. The method treats defects as disturbances in normal fabric texture and uses image reconstruction to identify abnormalities.

The research is particularly relevant to situations where collecting large labelled datasets is difficult. Its limitation is that the approach concentrates on image-based defect detection and does not provide an integrated mechanism for machine monitoring, production analytics, and operational optimization.

Paper 6: Nasim et al. (2024)

Title: *Fabric Defect Detection in Real World Manufacturing Using Deep Learning*

Nasim et al. developed a deep learning approach using a dataset collected directly from Chenab Textiles, representing real manufacturing conditions. The researchers compared YOLOv8, YOLOv5, and MobileNetV2 SSD FPN Lite. YOLOv8n achieved the best reported result with an mAP of 84.8%, followed by YOLOv5n at 84.5% and MobileNetSSD-FPNLite at 77.09%. The study is significant because it evaluates models using real industrial fabric images rather than relying only on laboratory datasets.

The study still focuses mainly on visual defect detection and does not combine defect information with machine-level production data or optimization decisions.

Paper 7: Guo et al. (2024)

Title: *Intelligent Quality Control of Surface Defects in Fabrics: A Comprehensive Research Progress*

Guo et al. reviewed intelligent fabric defect detection from the perspectives of algorithms, datasets, detection systems, performance, and industrial applicability. The study compares traditional

methods, machine learning, and deep learning approaches and discusses their strengths and weaknesses. The study highlights the importance of computer vision for intelligent textile manufacturing and identifies future research directions for fabric inspection. It also emphasizes issues related to datasets, detection performance, real-time operation, and applicability.

Paper 8: Carrilho et al. (2024)

Title: *Toward Automated Fabric Defect Detection: A Survey of Recent Computer Vision Approaches*

Carrilho et al. reviewed recent computer-vision approaches for automated fabric defect detection, with particular attention to machine-learning methods. The study explains that manual inspection remains common but can be costly and time-consuming, motivating automated inspection systems. The paper identifies automated defect detection as an important part of industrial quality control and discusses promising future directions for computer-vision-based textile inspection.

3. COMPARATIVE ANALYSIS

3.1 Comparative Literature Survey

Author/Year	Methodology	Dataset/Tools	Key Finding	Main Limitation
Kahraman & Durmuşoğlu, 2023	Deep learning review	Published fabric-defect studies	Deep learning is increasingly used for defect detection	Broad review; does not provide one integrated production system
Lin et al., 2023	Improved YOLOv5 + Swin Transformer	Tianchi, 10,321 images	Improved small-defect detection and real-time suitability	Mainly focused on visual inspection
Lin et al., 2023	AC-YOLOv5 + ASPP + CSE	Industrial self-built dataset	Reported 99.1% detection accuracy	Limited integration with production optimization

Szarski & Chauhan, 2022	Unsupervised anomaly detection	Dry carbon textile	Reduces dependence on labelled defect samples	Specific textile application
Hu et al., 2024	SA-Pix2pix + transfer learning	Fabric defect images	Addresses limited samples and complex patterns	Primarily image-based detection
Nasim et al., 2024	YOLOv8, YOLOv5, MobileNetSSD	Chenab Textiles dataset	YOLOv8n achieved 84.8% mAP	Limited production-system integration
Guo et al., 2024	Comprehensive review	Multiple datasets and methods	Identified algorithm, dataset, and system-level challenges	Review rather than implementation
Carrilho et al., 2024	Computer vision survey	Recent research	Identified progress and future directions	Focused mainly on defect detection

The comparison shows that YOLO-based methods provide a strong balance between detection accuracy and processing speed, while transfer-learning and unsupervised approaches can help address limited labelled data. However, the studies generally concentrate on the inspection component rather than integrating inspection results with production monitoring, root-cause analysis, optimization, and operational alerts.

4. CRITICAL ANALYSIS AND RESEARCH GAP

4.1 Limitations of Existing Approaches

The reviewed literature reveals several common limitations.

First, many studies depend on specific datasets collected under controlled or limited manufacturing conditions. This can reduce the ability of a model to generalize to different fabrics, lighting conditions, machines, and production environments. Recent research continues to identify dataset diversity and quality as major challenges.

Second, small and irregular defects remain difficult to detect. Fabric defects can have different sizes, shapes, aspect ratios, and appearances. Improved YOLO-based studies specifically address small-object detection because conventional approaches can miss small defects.

Third, complex fabric textures and patterned backgrounds can produce false detections or missed defects. Research on AC-YOLOv5 and transfer-learning approaches demonstrates the need for improved feature extraction and robustness in such environments.

Fourth, many existing studies evaluate defect detection as an isolated computer-vision problem. They do not connect defect information with machine conditions, production rates, downtime, or corrective actions.

4.2 Identified Research Gap

The major research gap identified from the literature is the lack of a unified smart textile manufacturing platform that combines:

Real-time defect detection + machine monitoring + production analytics + root-cause analysis + alerts + optimization

Existing research provides strong individual solutions for visual defect detection, but these solutions are generally not integrated into a complete production-management workflow.

Another important gap is the need for robust real-world operation across different fabric types and manufacturing conditions. Recent reviews continue to highlight challenges involving dataset size, imbalance, complex backgrounds, defect-scale variation, detection speed, and data acquisition.

Therefore, the proposed platform aims to move beyond detecting a defect and provide useful production information that can support timely decision-making.

5. FUTURE RESEARCH DIRECTIONS

5.1 Real-Time Edge-Based Inspection

Future systems can use lightweight deep learning models on edge devices located near production machines. This can reduce communication delays and support faster defect detection.

5.2 Larger and More Diverse Datasets

Future research should develop datasets containing different fabric materials, colors, patterns, defect types, lighting conditions, and machine configurations. This can improve model generalization.

5.3 AI and IoT Integration

Defect detection can be combined with IoT sensor data such as machine status, temperature, vibration, production speed, and downtime. Combining these data sources can help identify relationships between machine conditions and fabric defects.

5.4 Predictive Maintenance

Historical machine and defect data can be analyzed to identify patterns that indicate possible equipment failures. Predictive maintenance can reduce unexpected downtime and improve production reliability.

5.5 Automated Root-Cause Analysis

Future systems can correlate defect types with machine parameters, raw materials, environmental conditions, and production stages to identify likely causes automatically.

5.6 Production Optimization

AI-based recommendations can be developed to adjust machine parameters, production schedules, or maintenance activities based on observed defect and production trends.

5.7 Continuous Learning

Future systems can incorporate newly collected production data into model improvement workflows. This can help the defect detection model adapt to new fabrics and previously unseen defect patterns.

6. CONCLUSION

The literature survey demonstrates that deep learning and computer vision have become important technologies for automated textile defect detection. YOLO-based models have shown strong potential for real-time inspection, while transfer learning and unsupervised methods provide solutions for complex patterns and limited labelled data.

However, the reviewed research also shows that significant challenges remain in dataset diversity, small-defect detection, complex textile backgrounds, real-time operation, and generalization to real manufacturing conditions.

The major research gap is the limited integration between automated fabric inspection and broader manufacturing intelligence. The proposed Smart Textile Manufacturing Quality Inspection and Production Optimization Platform addresses this gap by combining defect detection with machine monitoring, production analytics, alerts, root-cause analysis, and optimization.

7. REFERENCES

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